

Decoding Behavior: Utilizing Virtual Reality Digital Marker and Machine Learning for Early Detection of Mild Cognitive Impairment

Yuwon Kim
Department of Applied Artificial
Intelligence, Seoul National
University of Science and Technology,
Republic of Korea
ferry@seoultech.ac.kr

Jinseok Park
Department of Neurology, College of
Medicine, Hanyang University, Seoul,
Republic of Korea
jinspark@hanyang.ac.kr

Hojin Choi
Department of Neurology, College of
Medicine, Hanyang University, Seoul,
Republic of Korea
chj@hanyang.ac.kr

Martin Loeser
Department of Computer Science,
Electrical Engineering and
Mechatronics, ZHAW Zurich
University of Applied Sciences,
Winterthur, Switzerland
martin.loeser@zhaw.ch

Hokyoung Ryu
Graduate School of Technology and
Innovation Management, Hanyang
University, Seoul, Republic of Korea
hryu@hanyang.ac.kr

Kyoungwon Seo
Department of Applied Artificial
Intelligence, Seoul National
University of Science and Technology,
Republic of Korea
kwseo@seoultech.ac.kr

ABSTRACT

The imperative for early mild cognitive impairment (MCI) detection is underscored by the limitations of traditional biomarkers, high cost and invasiveness, and they often fail to capture behavioral changes in MCI patients associated with impaired instrumental activities of daily living (IADL). This study introduces a cost-effective, non-invasive alternative using digital markers, “virtual kiosk test”, which involves performing IADL tasks such as ordering food via a kiosk in virtual reality (VR) to detect MCI at an early stage. Involving 20 healthy controls and 31 MCI patients, four key behavioral features within VR digital markers effectively differentiate groups: hand movement speed, proportion of fixation duration, time to completion, and the number of errors. A machine learning model demonstrated high effectiveness with 93.3% accuracy, 100% sensitivity, 83.3% specificity, 90% precision, and a 94.7% F1-score in group differentiation. Findings suggest that observing behaviors via the virtual kiosk test within 5 minutes can be an efficient approach for early MCI detection, acting as reliable VR digital markers.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Empirical studies in HCI**; **Interaction paradigms**; • **Virtual reality**; • **Computing methodologies**; • **Machine learning**; • **Machine learning approaches**; • **Kernel methods**; • **Support vector machines**;

KEYWORDS

Virtual reality, Digital marker, Machine learning, Behavior, Early detection, Mild cognitive impairment

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1 INTRODUCTION

As the global prevalence of Alzheimer’s disease (AD) continues to rise [1], the importance of early detection of mild cognitive impairment (MCI), a precursor stage to AD, becomes increasingly critical. MCI represents the last window of opportunity for potential cognitive recovery [2]. Traditional biomarkers such as cerebrospinal fluid [3] and magnetic resonance imaging [4] provide valuable diagnostic insights for MCI. However, their high costs and invasive nature limit their widespread use in ongoing clinical observation. In contrast, digital markers derived from behavioral data collected through digital devices like smartphones [5, 6] and tablets [7, 8] offer a more affordable and user-friendly alternative for assessing cognitive impairment. Therefore, the development and validation of digital markers as reliable detection tools represent an essential area of research.

Instrumental activities of daily living (IADL) tasks, which are complex daily activities requiring significant cognitive effort, are particularly promising for digital marker development [9-13]. It is observed that MCI patients often underperform in IADL tasks when compared to healthy controls [14, 15]. An example of such a task is ordering menu items from a kiosk, which requires memory to recall the desired item, attention to keep track of the ordering process, and executive function to organize the steps of the order sequentially [16]. Due to these reasons, our study leverages the

scenario of kiosk interaction to gather digital markers that could aid in the early detection of MCI.

The advancement of virtual reality (VR) technology allows for the collection of behavioral data from IADL tasks within a controlled setting, providing more ecologically valid data compared to methods employing mobile devices and tablets [17–22]. Previous research has shown the effectiveness of behavioral data derived from VR tasks, including hand movements [17, 23–25] and eye movements [26–28], as well as overall performance data [29, 30], in distinguishing between MCI patients and healthy older individuals. However, the challenge of effectively integrating these diverse behavioral data into digital markers for clinical use remains unresolved. Addressing this, our study focuses on developing and validating a “virtual kiosk test” as a VR digital marker for the early detection of MCI. This involves collecting real-time behavioral data and applying machine learning techniques for thorough analysis.

This study introduces a novel approach on two fronts: firstly, employing VR to aid in the early detection of MCI through IADL scenarios, and secondly, quantifying cognitive impairment by integrating behavioral data such as hand and eye movements with performance data. Overall, the deployment of this VR digital marker is expected to complement traditional biomarkers, offering frequent and ecologically sound data collection that is both cost-effective and non-invasive for MCI patients. The study is guided by three primary objectives:

- Identifying critical behavioral characteristics within the data collected during the virtual kiosk test, which distinctly separate MCI patients from healthy controls
- Investigating the relationship between outcomes of standard neuropsychological tests and those from the virtual kiosk test
- Developing and validating a machine learning model for the early detection of MCI, utilizing data from the virtual kiosk test, and analyzing its performance metrics

2 METHODS

2.1 Participants

In this research, we recruited 51 participants from Hanyang University Hospital, Republic of Korea from January 2022 to July 2023. The participant group comprised 20 healthy controls and 31 patients diagnosed with MCI. The healthy control group was selected from volunteers at a medical center. MCI patients were randomly chosen from general outpatients exhibiting decreased cognitive function during their annual wellness check-ups. The MCI diagnosis was conducted using the Seoul Neuropsychological Screening Battery-Core (SNSB-C), a validated screening tool for Korean population [31], by two experienced neurologists with 18 and 22 years of practice, respectively. They followed the diagnostic criteria established by Albert et al. [32], which encompassed (1) cognitive change, (2) impairment in one or more cognitive domains, (3) preservation of functional abilities, and (4) absence of dementia. The study set specific inclusion criteria: (1) participants should be over 50 years old, as per prior research [33, 34], to minimize age-related cognitive impairment biases [35], and (2) they must be capable of perceiving auditory and visual stimuli to interact within a virtual environment. The exclusion criteria included: (1) inability to read,

(2) a history of neurodegenerative or psychiatric disorders, and (3) any record of dementia or brain surgery. All participants were given a detailed explanation of the experiment and provided written informed consent. The study was designed and conducted in accordance with the guidelines and approval of the institutional review board of Hanyang University Hospital, Republic of Korea (HYUH-2021-08-020-004).

2.2 Neuropsychological Tests

To evaluate the cognitive functions of the participants, the study employed the SNSB-C. This battery includes five distinct neuropsychological tests that assess various cognitive functions. The SNSB-C encompasses: (1) Digit Span Test: Forward+Backward (DST: F+B) for measuring attention, (2) Short Form of the Korean-Boston Naming Test (S-K-BNT) for language function, (3) Rey Complex Figure Test (RCFT) for visuospatial function, (4) Seoul Verbal Learning Test-Elderly’s Version: Delayed Recall (SVLT-E: DR) for memory assessment, and (5) Korean-Trail Making Test-Elderly’s Version: Part B (K-TMT-E: B) for evaluating frontal-executive function (for a detailed description, refer to Table A1).

2.3 Virtual Kiosk Test

In the virtual kiosk test, participants’ behavior, including hand and eye movements and various performance data, was recorded while they completed an IADL task. This test is composed of ordering menu items using a virtual kiosk, structured to mitigate potential risks such as falls and cybersickness through a preliminary risk assessment. To minimize the risk of falls, the experiment was conducted within a confined space of 1.3 m by 1.3 m by 2 m, with participants seated. Additionally, to address cybersickness stemming from low-resolution VR, a high-resolution virtual kiosk test was developed and implemented. As depicted in Figure 1, participants wore an HTC VIVE Pro Eye head-mounted display and interacted with the high-quality virtual kiosk using a hand controller in their right hand. Two base stations tracked their hand movements. Inside the head-mounted display, embedded sensors monitored their eye movements every 100 ms from the start of the virtual kiosk test as they interacted with the virtual environment.

The virtual kiosk test consisted of six sequential steps: (1) selecting a place to eat, (2) choosing a burger item, (3) choosing a side item, (4) selecting a drink item, (5) selecting on a payment method, and (6) entering a 4-digit payment password (refer to the Appendices for a demonstration video). Before beginning the virtual kiosk test, participants were given these instructions: “The place to eat is a restaurant. Please order a shrimp burger, cheese sticks, and a Coca-Cola using the kiosk. Please use a credit card as the payment method. The card payment password is 6289.”

During the virtual kiosk test, behavioral data on hand and eye movements and other performance data were gathered. Four key behavioral features were identified: (1) hand movement speed [23, 36], (2) proportion of fixation duration [37, 38], (3) time to completion [23], and (4) the number of errors [23, 39]. Table 1 provides a detailed overview of these features extracted from the virtual kiosk test.

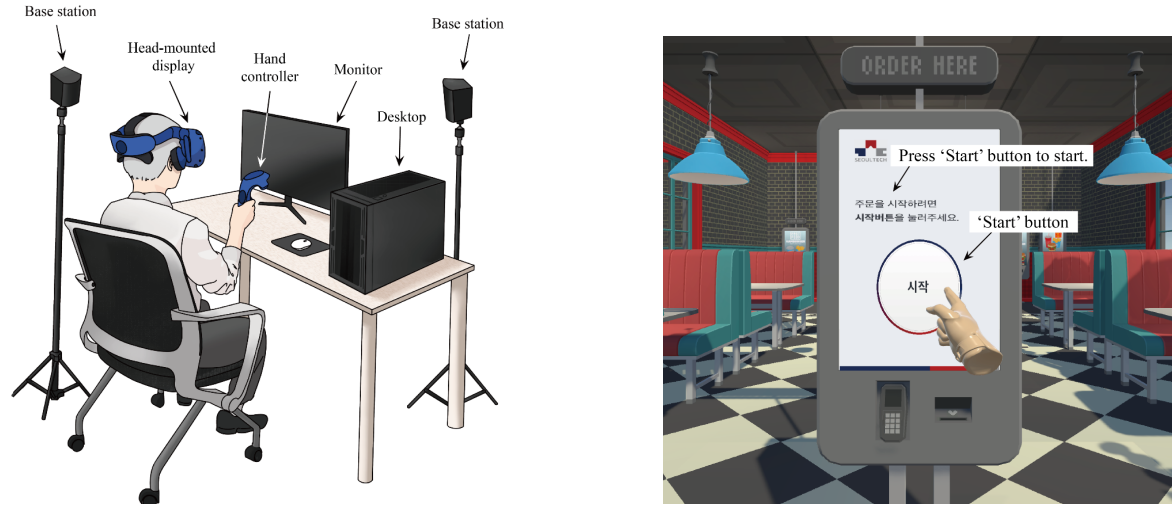


Figure 1: Setup for the virtual kiosk test (left) and participant's view of the virtual kiosk (right).

Table 1: Four key behavioral features extracted from the virtual kiosk test

Virtual kiosk test	Description
Hand movements	
Hand movement speed (m/s)	The mean speed of hand movements recorded during a participant's engagement with the virtual kiosk test
Eye movements	
Proportion of fixation duration (%)	The proportion of time a participant spent concentrating on the instructed menu item out of all other menu items
Performance data	
Time to completion (seconds)	The cumulative duration taken by a participant to finish the virtual kiosk test
The number of errors	The overall count of errors recorded during a participant's completion of the virtual kiosk test

2.4 Procedures

Each participant underwent both the neuropsychological tests and the virtual kiosk test in a counterbalanced order, under the supervision of neurologists responsible for diagnosing MCI. The neuropsychological tests took around 40 minutes, whereas the virtual kiosk test was completed in about five minutes. Before starting the virtual kiosk test, participants went through an eye calibration session and participated in two practice sessions to ensure they could successfully complete the entire test process. These practice sessions typically lasted 2-3 minutes but could extend to a maximum of five minutes. Including the eye calibration and practice, this preparatory phase totaled about 10 minutes. While the option to stop the experiment due to cybersickness was available, no participant expressed a desire to do so. Overall, the complete experimental procedure took approximately 55 minutes.

2.5 Analysis

To evaluate the effectiveness of the virtual kiosk test as a VR digital marker for early MCI detection, a series of statistical tests were performed using SPSS Statistics 27 software. We began by applying

chi-squared tests and independent-sample t-tests to compare basic demographic characteristics and neuropsychological test results between healthy controls and MCI patients. This was followed by additional independent-sample t-tests to assess differences in the virtual kiosk test results between the two groups. A multiple regression analysis was then conducted to investigate the correlation between neuropsychological test results and virtual kiosk test results. To address the issue of multiple comparisons, all statistical analyses were adjusted using the Bonferroni correction method.

Building on these findings, we selected key behavioral features from the virtual kiosk test to develop a machine learning model, aiming to differentiate between healthy controls and MCI patients. To reduce the risk of overfitting, we employed a stratified train-test split in a 7:3 ratio. The model's training and deployment were carried out using Python 3 and the Scikit-learn library. The performance of this machine learning model was evaluated based on accuracy, sensitivity, specificity, F1-score, and the area under the receiver operating characteristic curve (AUC).

Table 2: Comparative analysis of basic demographic characteristics and neuropsychological test results between healthy controls and MCI patients

Participants characteristics	Healthy controls (N = 20)	MCI patients (N = 31)	<i>p</i> value ^a
Basic demographic characteristics			
Gender (female)	10	16	> .99
Age (years)	70.95 ± 6.02	72.68 ± 7.75	> .99
Educational level (years)	12.60 ± 4.54	9.32 ± 4.98	.16
Neuropsychological test results			
DST: F+B (number of correct answers)	10.60 ± 1.54	8.52 ± 1.82	< .001
S-K-BNT (number of correct answers)	13.05 ± 1.43	11.0 ± 1.88	< .001
RCFT (score)	33.28 ± 2.44	26.42 ± 6.56	< .001
SVLT-E: DR (number of correct answers)	7.15 ± 2.30	2.52 ± 2.59	< .001
K-TMT-E: B (time to completion)	38.75 ± 22.93	114.65 ± 103.85	.02

^a Adjusted *p* values after the Bonferroni correction (.05/8 = .006).

Table 3: Comparative analysis of four behavioral features from the virtual kiosk test between healthy controls and MCI patients

Participants characteristics	Healthy controls (N = 20)	MCI patients (N = 31)	<i>p</i> value ^a
Hand movements			
Hand movement speed (m/s)	0.23 ± 0.06	0.17 ± 0.05	.004
Eye movements			
Proportion of fixation duration (%)	56.0 ± 13.1	43.7 ± 17.5	.04
Performance data			
Time to completion (s)	40.41 ± 19.35	104.95 ± 82.19	.004
The number of errors	1.60 ± 1.60	4.16 ± 2.75	.001

^a Adjusted *p* values after the Bonferroni correction (.05/4 = .012).

3 RESULTS

3.1 Basic Demographic Characteristics and Neuropsychological Test Results

Table 2 illustrates that, based on chi-squared tests and independent-sample *t*-tests, there were no notable differences in the basic demographic characteristics between the healthy controls and MCI patients. Nonetheless, significant disparities were observed in the neuropsychological test results between these two groups.

3.2 Differences in Virtual Kiosk Test Results between Healthy Controls and MCI Patients

Table 3 demonstrates that the independent-sample *t*-tests identified significant differences between healthy controls and MCI patients in all four behavioral features of the virtual kiosk test. These features include hand movement speed, proportion of fixation duration, time to completion, and the number of errors. Specifically, compared to healthy controls, MCI patients showed significantly slower hand movement speed ($t_{49} = 3.45$; $p = .004$), reduced proportion of fixation duration ($t_{49} = 2.69$; $p = .04$), an increased time to completion ($t_{49} = -3.44$; $p = .004$), and a higher number of errors ($t_{49} = -3.77$; $p = .001$) during the virtual kiosk test.

3.3 Correlation between Neuropsychological Test Results and Virtual Kiosk Test Results

A multiple regression analysis was performed to explore the relationship between the four key behavioral features obtained from the virtual kiosk test and the results of neuropsychological tests (detailed in Table A2 in the Appendices). The analysis revealed that *hand movement speed* significantly correlated with several cognitive functions: attention (DST: F+B; $r = 0.36$, $p = .047$), language function (S-K-BNT; $r = 0.46$, $p = .003$), visuospatial function (RCFT; $r = 0.43$, $p = .007$), and frontal-executive function (K-TMT-E: B; $r = -0.42$, $p = .01$). The *proportion of fixation duration* was notably linked with attention (DST: F+B; $r = 0.53$, $p = <.001$), visuospatial function (RCFT; $r = 0.51$, $p = <.001$), memory (SVLT-E: DR; $r = 0.43$, $p = .008$), and frontal-executive function (K-TMT-E: B; $r = -0.41$, $p = .01$). *Time to completion* showed significant correlations with attention (DST: F+B; $r = -0.40$, $p = .02$), visuospatial function (RCFT; $r = -0.49$, $p = .001$), and frontal-executive function (K-TMT-E: B; $r = 0.46$, $p = .003$). Finally, *the number of errors* was significantly correlated with all five cognitive domains: attention (DST: F+B; $r = -0.45$, $p = .004$), language function (S-K-BNT; $r = -0.53$, $p = <.001$), visuospatial function (RCFT; $r = -0.62$, $p = <.001$), memory (SVLT-E: DR; $r = -0.41$, $p = .01$), and frontal-executive function (K-TMT-E: B; $r = 0.57$, $p = <.001$).

3.4 Machine Learning Classification Performance using Virtual Kiosk Test Results

To refine the early detection process for MCI, a machine learning model was developed using normalized behavioral features from the virtual kiosk test: hand movement speed, proportion of fixation duration, time to completion, and the number of errors. The normalization process scaled these features within a 0 to 1 range to ensure consistent model performance. A support vector machine (SVM) was chosen as the most effective model based on various performance metrics, including accuracy, sensitivity, specificity, precision, F1-score, and AUC, as detailed in Table A3 in the Appendices. The optimal SVM parameters were established using a grid search, selecting a radial basis function kernel with $C = 1.0$ and $\gamma = 1.86$. Leave-one-out cross-validation was implemented to avoid overfitting.

The SVM model demonstrated excellent classification performance between healthy controls and MCI patients when it integrated all four behavioral features from the virtual kiosk test, as depicted in Figure 2. The results showed 93.3% accuracy, 100% sensitivity, 83.3% specificity, 90% precision, 94.7% F1-score, and an AUC of 0.98. Models trained with a single feature experienced a significant drop in performance. Specifically, the model using only hand movement speed achieved 80% accuracy, 88.9% sensitivity, 66.7% specificity, 80% precision, 84.2% F1-score, and an AUC of 0.94. The model based solely on proportion of fixation duration was next, with 60% accuracy, 100% sensitivity, 0% specificity, 60% precision, 75% F1-score, and an AUC of 0.93. The model that used time to completion as the single feature had 80% accuracy, 77.8% sensitivity, 83.3% specificity, 87.5% precision, 82.4% F1-score, and an AUC of 0.89. Lastly, employing the number of errors alone resulted in 73.3% accuracy, 88.9% sensitivity, 50% specificity, 72.7% precision, 80% F1-score, and an AUC of 0.88.

4 DISCUSSION

4.1 Principal Findings

The results of this research endorse the virtual kiosk test as an effective VR digital marker for the early detection of MCI, utilizing behavioral data gathered from VR-implemented IADL tasks. The study identified distinct behavioral discrepancies between healthy controls and MCI patients when interacting with the virtual kiosk, specifically in terms of slower hand movement speed, decreased proportion of fixation duration, prolonged time to completion, and increased number of errors among MCI patients. An SVM model, trained on these behavioral features, achieved outstanding discrimination accuracy, with a 93.3% correct classification rate, 100% sensitivity, 83.3% specificity, 90% precision, 94.7% F1-score, and an AUC of 0.98, validating its use for early MCI detection. Notably, this performance surpassed that of previous studies employing non-invasive markers to train machine learning models for early MCI detection. For instance, Schäfer et al. [40] achieved an AUC of 0.81 using speech data, while Ghoraani et al. [41] reported 88% classification accuracy utilizing gait data. Furthermore, our analysis unveiled significant correlations between the behavioral measures from the virtual kiosk test and conventional neuropsychological

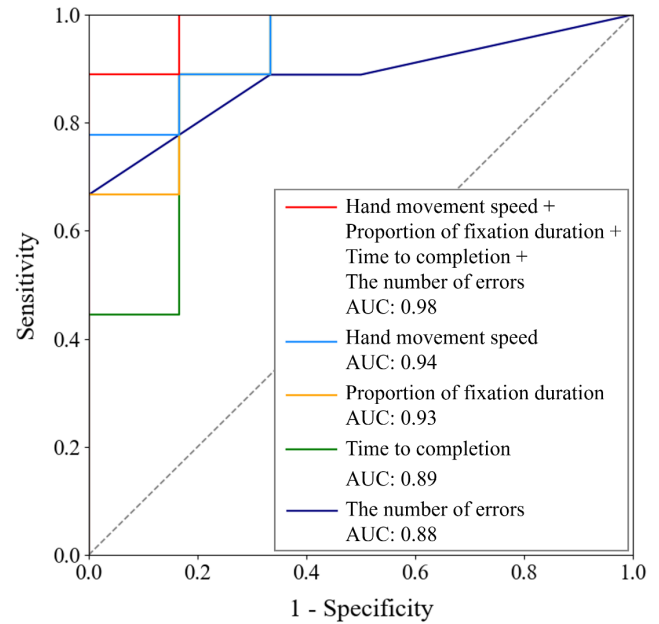


Figure 2: Receiver operating characteristic curve for SVM models based on key virtual kiosk test features.

test results, reinforcing the real-world applicability and ecological validity of the virtual kiosk test's behavioral data.

The effectiveness of using hand and eye movement analysis in VR as a means for early MCI detection can be understood within the framework of embodied cognition theories [42-44]. These theories suggest that our cognitive processes are deeply interconnected with our physical movements [45]. Analyzing the coordination of cognitive function and physical action can provide deeper insights into the early markers of neurodegenerative diseases. Observing hand and eye movements while participants complete a cognitively intensive IADL task, such as selecting items on a virtual kiosk, offers valuable information about the interconnectedness of cognition and action. For example, Figure 3 shows hand movement trajectories among healthy controls and MCI patients. MCI patients' trajectories, particularly in steps 3 to 6, show more intricate patterns in menu selection, potentially indicative of cognitive deficits affecting attention, language, visuospatial abilities, and frontal-executive functions. For instance, a lack of focused attention may lead to irregular hand and eye movements and indecisive actions. Figure 4 highlights unique eye movement patterns in MCI patients, who often get distracted by non-relevant menu items, a symptom tied to deficits in various cognitive domains. Dispersed eye fixations during the test might reveal spatial processing impairments, leading to unfocused gaze and attention shifts to unrelated items. These findings underscore the importance of analyzing motor actions in a VR setting to enhance the process of early MCI detection.

Employing VR digital markers from the virtual kiosk test offers tangible benefits for MCI detection, enhancing traditional biomarkers and other digital approaches. Studies show that VR digital markers align closely with the sensitivity and specificity of established

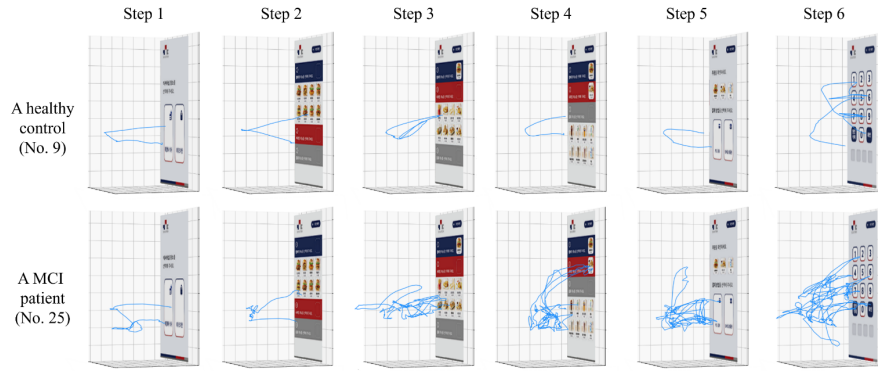


Figure 3: Comparison of hand movement trajectories between a healthy control and an MCI patient.

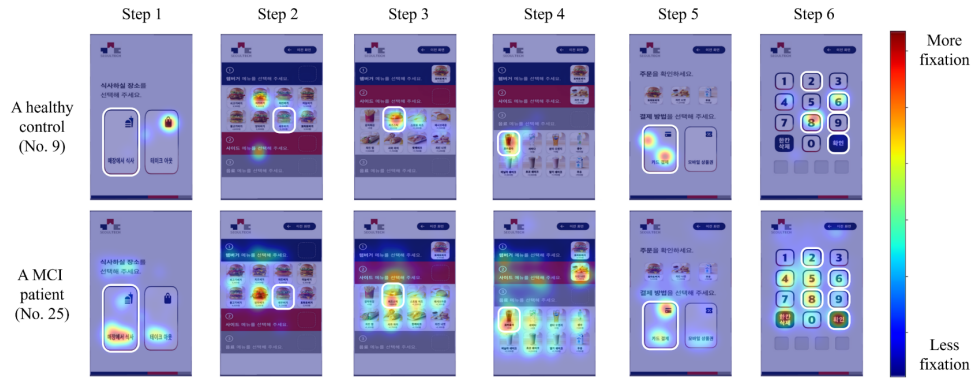


Figure 4: Heatmap analysis of eye movements in a healthy control and an MCI patient.

biomarkers [46, 47], yet they stand out for their cost-effectiveness and non-invasive nature, allowing for more frequent and less burdensome data collection [48, 49]. The virtual kiosk test boasts a swift completion time of only 5 minutes, a stark contrast to the lengthy 40-plus minutes typically needed for neuropsychological testing. Despite focusing on a single IADL task, our study yielded strong classification results, reflecting the depth of cognitive engagement required for the kiosk interaction. Participants also reported enjoyment and interest in repeating the test, indicating its potential role as a cognitive rehabilitation tool for MCI individuals [50–54]. While promising, larger-scale studies are needed to further substantiate these findings. Ultimately, VR digital markers from the virtual kiosk test present as valuable additions to the suite of tools for early MCI detection.

4.2 Limitations

This study has several limitations. Firstly, the small sample size could potentially bias the machine learning model, limiting its wider applicability. Future studies should look to increase the sample size and extend data collection through collaborations across multiple institutions to address this concern. Secondly, while the study made strides in correlating virtual kiosk test performances with neuropsychological assessments, there remains a need for further

research to fully understand the scope of digital markers in relation to physiological and anatomical data, such as electroencephalography [55] and magnetic resonance imaging [56–58]. Finally, the current study focused on using behavioral data within a machine learning framework; however, forthcoming research might explore the application of deep learning techniques, which could provide a more nuanced analysis of hand and eye movement patterns in MCI patients from both spatial and temporal dimensions.

4.3 Conclusions

This study provides initial evidence that behavioral features from the virtual kiosk test hold considerable promise as digital markers for the early detection of MCI. Using an SVM model, key behavioral features successfully differentiated between healthy controls and MCI patients, achieving excellent classification performance with 93.3% accuracy, 100% sensitivity, 83.3% specificity, 90% precision, 94.7% F1-score, and an AUC of 0.98. These behavioral features also correlated strongly with various cognitive functions, supporting their relevance in cognitive assessments. The results highlight the potential of using VR digital markers, particularly those analyzing hand and eye movements within a VR context, for early detection of MCI, thus offering implications for further research in MCI rehabilitation.

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